

Problem solving with algebra



- 1 Adam sells chocolates to raise money for charity. Each chocolate costs \$6.
 - a If Adam sells q chocolates, write an algebraic expression for the amount of money he raises.
 - b How much do 4 chocolates cost?
 - c How much do 5 chocolates cost?
 - d Is it possible to raise exactly \$13 if Adam sells chocolates at \$6?

- 2 Valentina's hens can produce 5 eggs each day.
 - a If Valentina collects the eggs from her hens for y days, write an algebraic expression for the total number of eggs.
 - b Find the number of eggs Valentina will have after 30 days.

- 3 John breeds dogs for sale. Each of his d dogs gives birth to p puppies.
 - a Write an algebraic expression for the total number of puppies.
 - b Can the value of p be a fraction?
 - c If he has 3 dogs, find the total number of puppies.

- 4 Judy has 72 pencils, which she shares evenly among the students in her class.
 - a If there are q students in her class, write an algebraic expression for the number of pencils given to each student.
 - b If there are 18 students in the class, how many pencils does each student receive?
 - c If there are 36 students in the class, how many pencils does each student receive?
 - d As the number of students in the class increases, does the number of pencils each student receives increase or decrease?

- 5 At John's birthday party, cake slices are shared evenly among his 8 guests.
 - a If John shares x slices, write an algebraic expression for the number of slices given to each guest.
 - b If John shares 32 cake slices, find the number of slices given to each guest.

- 6 Kenneth uses a watering can to give each of his plants an equal amount of water.
 - a If his watering can contains m mL of water and he has n plants, write an algebraic expression for how much water is given to each plant.



- 7** Robert visits a carnival that costs \$5 to enter, and each ride costs \$1 per person.
- a** If Robert decides to go on b rides, write an algebraic expression for the total amount he spends at the carnival.
 - b** If Robert goes on 6 rides, calculate the amount of money he spend in total.
- 8** Ben has 241 GB of available space on an external hard drive. He decides to transfer his files to this drive.
- a** If the transferred files have a total size of x GB , write an algebraic expression for the amount of space left on the external drive.
 - b** Find the space left if Ben transfers 16 GB in total.
- 9** To get to school, Amelia walks for 9 minutes to the bus stop and waits 2 minutes for the bus to arrive. She rides the bus for the rest of the way to school.
- a** If the bus trip takes n minutes, write an algebraic expression for the total time it takes Amelia to get to school.
 - b** Yesterday the bus trip took 15 minutes. Find the total time it took for Amelia to get to school yesterday.
- 10** In a Year 7 class, students are either twelve years old or thirteen years old.
- a** If there are k twelve year-old and m thirteen year-old students, write an algebraic expression for the total number of students in the class.
 - b** If there are 4 twelve year-old and 16 thirteen year-old students, find the number of students are in the class.
- 11** Tennis coach Luigi has 5 balls left. To ensure he has enough for the next training session, he orders one new pack of balls for each of his y students.
- a** If the new tennis balls come in packs of 4, write an algebraic expression for the total number of tennis balls Luigi will have.
 - b** Find the number of tennis balls Luigi will have if he coaches 7 students.
- 12** Sarah has 14 cookies in her cookie jar. Her 2 children each eat a cookie every day.
- a** Write an algebraic expression for the number of cookies left in the jar after y days.
 - b** Are the number of cookies decreasing by the same amount each day?

- 13 Vanessa has \$700 in her bank account. She  uses the account to pay her mobile phone bill each month.
- If each monthly bill is \$14, write an algebraic expression for how much Vanessa has in her account after c months.
 - Find the amount of money she will have in her account after she pays her bill for 4 months.
- 14 Sally attends a dessert festival. Each ice cream costs \$6 and each milkshake costs \$8.
- If Sally buys c ice creams and m milkshakes, write an algebraic expression for the total cost.
 - How much would two ice creams and one milkshake cost?
 - Using algebra, how much would c ice creams and no milkshakes cost?
- 15 Laura has a piggy bank in which she collects $20c$ and $50c$ coins. After some time, she loses track of how many coins are in the piggy bank.
- Let m represent the number of $20c$ coins and n represent the number of $50c$ coins. Write an algebraic expression for the total value of Laura's coins in cents.
 - Laura breaks her piggy bank and discovers that she has thirteen $20c$ and twenty-seven $50c$ coins. Find the total value of these coins in dollars.
- 16 Valerie places a bird feeder in her garden. That day, she sees 3 birds use the feeder. The next day she sees 6 birds, and on the third day she sees 9 birds.
- If the number of birds continues to follow the pattern, find the number of birds that Valerie will see on the fourth day.
 - If the pattern continues, write an algebraic expression for the number of birds Valerie sees on the x th day.
- 17 Water is dripping from a tap into a large bucket, so that:
- After 1 hour, the water level in the bucket is 5 cm
 - After 2 hours the water level is at 10 cm
 - After 3 hours the water level reaches 15 cm
- If the tap stops leaking after a hours, write an algebraic expression for the water level in the bucket at this time.
 - Is the water level increasing by the same amount each hour?

18 Fred likes to go kayaking. He takes his boat down to the lake which is 400 m away from his house.



- After 1 minute of paddling he is 490 m away from his house
 - After 2 minutes of paddling he is 580 m away from his house
 - After 3 minutes of paddling he is 670 m away from his house
- a Fred paddles for n minutes in total. Write an algebraic expression for the distance from his house at this time.
- b How far does Fred travel in the kayak per minute?
- c Is Fred paddling at a constant pace for the first n minutes?
- d If Fred kayaks for 10 minutes, how far does he travel from his house altogether?

19 Kathleen cuts squares from 1 cm grid paper:

- The first square is $2\text{ cm} \times 2\text{ cm}$ and has an area that contains four 1 cm^2 pieces
 - The second square is $3\text{ cm} \times 3\text{ cm}$ and has an area that contains nine 1 cm^2 pieces
 - The third square is $4\text{ cm} \times 4\text{ cm}$ and has an area that contains sixteen 1 cm^2 pieces
- a If the pattern continues, find the number of 1 cm^2 pieces that fit in a square that is $5\text{ cm} \times 5\text{ cm}$.
- b Write an algebraic expression for the number of 1 cm^2 are in a square with a side length of $h\text{ cm}$.